Thesis Title

Thesis Subtitle



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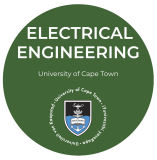
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October 2026

BSc (Eng) final year project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and Computer Engineering in the Department of Electrical Engineering at the University of Cape Town

Keywords: keyword1; keyword2; keyword3



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Wed 7 October, 2026

Abstract

Thesis Title

Joab Gray **Kloppers**

Wed 7 October, 2026

Acknowledgements

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List of Symbols

Symbol	Description	Unit
$\boldsymbol{B}$	Magnetic flux density	T
$\boldsymbol{E}$	Electric field	V/m
$\boldsymbol{J}$	Current density	A/m^2
t	Time	s
ε_0	Permittivity of free space (8.85×10^{-12})	$C^2/N{\cdot}m^2$
μ_0	Permeability of free space ($4\pi \times 10^{-7}$)	N/A^2

Chapter 1

Introduction

1.1 Background

1.1.1 Biological Activity

Sand prawn activity has a large effect on the surrounding ecosystem. Pillay and Branch [2] explain that by physically burrowing into the sediment, they distribute buried sediment to the surface, alter the surface structure via the introduced mounds, which can increase surface area, and alter both chemical and structural properties of the sediment through burrowing and nutrient mixing. Venter et al [3] further shows that sand prawn bi-directional water pumping filters can reduce phytoplankton biomass by up to 70% in estuarine environments which can help with ecosystem resilience.

1.1.2 Technical application

Schenone et al [1] demonstrates that surface microtopography can be used as a reliable metric to predict biochemical processes. However common metrics for measuring surface roughness characteristics such as (list them here) are often invasive and result in the damage of the very structures under measurement. Azhar et al [4] demonstrated that the use of low cost RGB-D cameras can be used to produce depth images of the sediment surface, which can then be processed to analyse the surface properties.

Sub-subsection

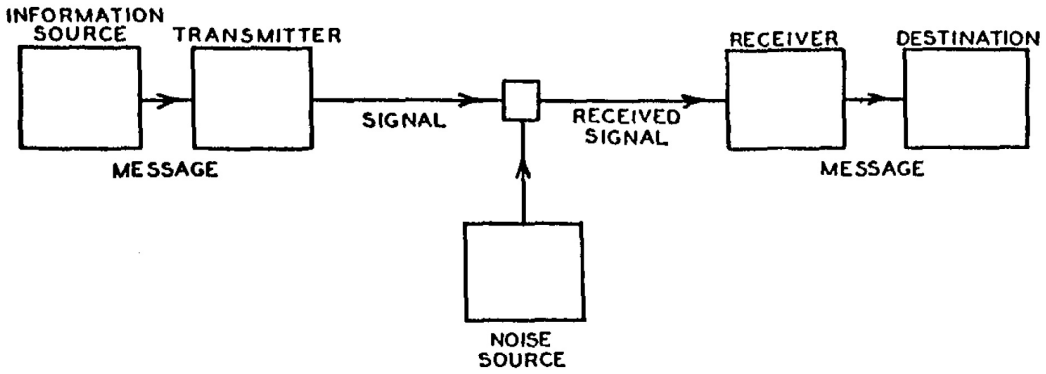


Figure 1.1: A simplified block diagram of a general communication system. A communication system can be abstracted as follows: an *information source* produces a message to be sent through a *transmitter* which converts it to a signal which can be sent via a communication *channel*. The received signal is noisy and is converted back to the message by the *receiver* which is then delivered to the *destination*. Taken from shannon1948

Table 1.1: Table title. Extended description of tables contents.

Heading 1	**Heading 2**	
	Subheading 1 [units]	*Subheading 2* [units]

1.2 Problem statement

Equations are numbered and include the chapter number. As they are considered part of the text, they include punctuation, and each parameter must be described in the text below. Ensure that the units are stated and where possible use SI units. For example, the Maxwell equations are expressed as:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}, \tag{1.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1.2}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{1.3}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}, \tag{1.4}$$

where $\boldsymbol{B}$ is the magnetic flux density in T, $\boldsymbol{J}$ is the total current density in A/m^2, $\boldsymbol{E}$ is the electric field in V/m, μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), and ϵ_0 is the permittivity of free space (8.85 $\times 10^{-12}$ C^2/N·m^2).

1.3 Project objectives

The Global Positioning System (GPS) is used for positioning. The Marginal Ice Zone (MIZ) is of particular interest in the Southern Ocean (SO).

1.4 Scope, limitations and assumptions

1.5 Plan of development

Chapter 2

Literature review

2.1 Sandprawn Bioturbation and Ecological Significance

Pillay and Branch [2] describe how sand prawn species are prominent members in the benthic ecosystem. The sand prawns continuously change the sediment properties such as surface topography porosity and biochemistry, through actions such as burrowing and water pumping. These actions redistribute buried sediment to the surface which results in surface mounds which increase the effective surface area of the sediment by up to 15 times. Kristensen et al [5] further explains how thalassinid shrimp populations can turn up to 12kg of sediment per square meter per day. These mounds can increase the effective hydrodynamic resistance which can help to stagnate the water flow close to the sediment. While the paper mentions measurements specific to sand prawn species located in other regions, they provide an insight into approximate feature sizes that can be expected, for example *Callianassa subterranea* present in the North sea can produce mounds 5.4cm tall and 11.7cm wide, and *Pestarella candida*, located in the Mediterranean and Adriatic Seas, which can produce mounds 3cm tall and 15cm wide.

Venter et al [3] further explains how sand prawns, *Callianassa kraussi*, present in the South African coastlines and at the Langebaan Lagoon, function as prominent water filterers. Within the burrows they actively irrigate the sediment through bi-directional pumping of water to replenish oxygen levels within the burrows, and to remove waste products. This irrigation has a secondary effect where phytoplankton are filtered out of the water by the burrow walls. This filtering can help improve ecosystem resilience to harmful conditions such as nutrient pollution which promote phytoplankton blooms. In relation to this project, the papers methodology entails manual visual counting of burrow openings.

The next question with regards to biological activity, is weather surface roughness, specifically rugosity, is a valid measurement to gauge the underlying biological activity within the benthic ecosystem. To this, Schenone et al [1, 6] determined experimentally that analysing two dimensional surface rugosity, they were able to correlate surface roughness measurements with sub-surface activity. One paper [1] described the testing of a controlled laboratory environment to simulate the benthic ecosystem, which can be seen in Figure 2.1. They found that the simple two dimensional rugosity estimate managed to account for up to 61% of nutrient variability. The second paper[6] describes a more complex implementation, with the use of a single diver operated camera recording video of the sediment surface, and the use of visual based structure from motion (SFM) photogrammetry to obtain a three dimensional representation of the surface. similarly, the find that sediment microtopography explains up to 55% of species richness variance, which is of particular interest with regards to this project. These papers show that surface rugosity is a useful measurement when gauging biological activity, and they provide a useful insight into the computational pipeline which is used to convert from spatial meshes to elevation matrices, however there are considerations that need to be considered, these papers discuss fully submerged environments, thus the requirement of under water cameras and SFM, whilst this project focuses on tidally exposed surfaces, where fixed clear observations can be made with the use of a rig.

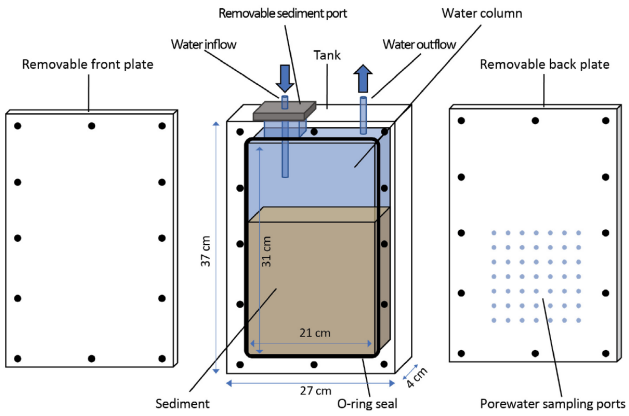


Figure 2.1: Custom incubation and flow through chamber by Schenone et al [1]

2.2 Surface Microtopography as a Proxy for Bioturbation

2.3 RGB-D Sensing for Non-Contact Surface Measurement

2.4 Summary and identified Research Gap

With the understanding that surface roughness statistics can provide a feasible proxy for benthic activity, there is a need for a method to scan sediment surfaces and extract topographical information. The current project, under the biological side **<PLEASE CHANGE THIS, SOMEHOW TIE IN THE EXISTING PROJECT>**, takes surface scans of static, exposed sediment, within the inter-tidal region in the Langebaan lagoon. These conditions mean that there is no need for submersion rated hardware, operated under dynamic conditions.

The current project is based off the work developed by Azhar et al [4]. The authors develop a framework that allows for extracting a millimetre scale 3D surface scans non-invasively, and at lower price scales than expensive LiDAR based systems. The system makes use of an off the shelf Asus Xiton pro Live 3D RGB-D structured light camera, mounted within an opaque plastic container to block infrared interference from sunlight. They then tested four polynomial de-trending algorithms to remove natural sediment gradients, and isolate the bioturbation. Due to the nature of the Asus Xiton, which makes use of structured light projection and a single IR camera, the system suffers when the surface is exposed to sunlight. Therefore the authors made use of an opaque plastic enclosure which is set on the sediment for scanning, however this limits the spatial resolution to only $0.3m^2$ with an effective camera resolution of only 240 x 200 pixels. Additionally the authors found that the higher order polynomial de-trending algorithms were sensitive to elevations at the edges of the image frame, requiring them to use a planar model which proved to be more robust.

Halmetschlager-Funek et al [7] and Rustler et al [8] provide useful insight by comparing several types of commonly available depth sensing cameras, encompassing different technologies such as structured light, active and passive stereo vision and time of flight (ToF). Halmetschlager-Funek et al [7] compares ten different sensors, with varying technologies and shows that sensor technologies like structured light and ToF generally struggle when exposed to high ambient light or reflective materials. These make it impractical to apply in the exposed environmental conditions considered for this project, as there is often direct sunlight, and small water pools which can increase reflectivity of the sediment. Rustler et al [8] compares four different stereo vision cameras, and finds that the Intel RealSense D435 is the superior camera for applications what are within one meter, however distance bias against planar surfaces increases significantly, compared to similar sensors such as the Intel RealSense D455, past 300cm distance. The author finds that for longer distance, the ZED 2 or Intel RealSense D455 are the best performers, with the main discerning factor being the image processing, as the ZED 2 requires dedicated GPU hardware for image processing, whereas the D455 does most of the processing on-board with some post precessing offloaded to the connected device.

Nguyen et al [9] discusses a method of deriving the lateral and axial noise profiles of the Microsoft Kinect using through analysis of specific scenes. To characterise lateral noise the authors make use of the fact the pixel statistical distributions along a depth edge are the same as statistical distribution of the point spreading factor (PSF), and thus can obtain a characterisation

on how points get blurred through the optics of the camera. For axial noise profiles, they analyse depth distributions across a plane of best fit, across various angles. Using these noise profiles, they are able to implement sensor specific filters that can filter noisy depth readings, whilst preserving geometric features. The benefit to this method is that it can be implemented across different sensors, enabling the application on the Intel RealSense D456 for improved depth filtering.

Metrological and critical characteriseation of intel d415 **<DONT THINK SHOULD USE THIS ONE>**

A benefit of using the Intel RealSense RGB-D cameras is the advanced ASIC which does all depth processing on device, thus eliminating the need for high performance computers during scans. Additionally the software development kit (SDK) for the RealSense family provides useful tools to the developer to tweak settings for specific applications, and to apply post-processing to scans for improved performance. Keselman et al, and Grunnet-Jepsen and Tong [10, 11] provide useful insight into the operation and theory of the Intel RealSense r200 and D400 series cameras respectfully. Keselman et al [10] describes how the correlation algorithm performs well under texture sparse conditions, and is more robust to noise compared to similar algorithms. The author also explains the limitations of the sensor, such as the minimum depth is dictated by the limited disparity search range, and maximum distance is limited by signal to noise ratio. Additionally, the use of a laser based feature projector can introduce some artifacting due to the effect of laser speckle noise. Grunnet-Jepsen and Tong [11] provide an outline on the post-processing methods provided by the Intel RealsSense SDK. This white paper explains how some of the processes work, and gives a general outline on the pipeline that should be applied to achieve a baseline filtered depth map.

One fundamental requirement of stereo vision is the need for textured surfaces, in order to perform disparity matching for depth computation. This need results as a reduction in performance on texture sparse environments. The Intel RealSense family address this with the use of a laser based IR projector which project semi random patterns onto the surface to artificially introduce visual texture. Grunnet-Jepsen et al [12] describes how the illumination power of the internal projector can be optimised for different light environments, and further how an external projector or dot pattern can be overlaid onto the surface to improve performance on texture sparse environments, as the D400 series cameras do not require a predefined pattern. These projected patterns can improve performance, as the achievable dot density is far higher than what the on board projector can achieve, and does not suffer with the same laser speckle problem.

<LOOK FOR A PAPER DISCUSSING TEXTURE OF WATER LOGGED SAND>

Intel Realsense camera depth testing methodology

pixel level extrinsics self calib??????

Chapter 3

Theoretical Backbone

3.1 Stereo Vision

3.2 Capture Alignment

3.3 Fusion

3.4 Evaluation

3.5 Roughness Calculation

3.6 Feature Detection

Chapter 4

System Design

4.1 Requirements Definition

4.2 Hardware Rig

4.3 Software Architecture and Processing Pipeline

Chapter 5

Intel RealSense Characterisation

Before any action is taken on developing the capture and processing pipeline for extracting surface roughness metrics, an in depth characterisation on the Intel RealSense D456 is required. This characterisation will inform decisions about the experimental rig, as well as the configuration of the camera itself for optimal scanning, considering the conditions experienced in situ.

Characterisation was performed in 2 main steps. Firstly, the RealSense camera was characterised with the use of pre-made software support for plane surface characterisation using the RealSense SDK Depth Quality Tool. These tools provide useful metrics for determining camera performance against smooth, flat planar surfaces. These metrics give an indication for what conditions might be optimal for the camera, such as height above surface, lighting conditions and tilt angle.

Using the results for planar surfaces as a baseline, next, the camera was characterised against known reference 3D objects shown in <**APENDIX**>. With these objects, a more focused sweep of which camera configuration settings result in the best reconstruction performance against these known objects.

5.1 Plane Surface Characterisation

As mentioned earlier, the first set of characterisation was performed against planar targets. with the use of metrics provided by the RealSense SDK. The first set of testing involved characterising the camera at different axial distances to the surface.

The depth quality tool provided by Intel recommends performing analysis against a plane, white, featureless wall. This results in the camera relying on its own IR laser projection for measuring depth. For all planar characterisation, custom c++ code was written to use the RealSense SDK provided by Intel. The custom code would allow the user to enter details about the scan, such as distance, angles, light intensity, and would capture 120 frames. The code would then set a dynamic region of interest (ROI) that would remain constant at a user set dimension of 50x50cm. The code would then make use of the SDK provided tools for fitting a plane to the scanned surface, and measuring the absolute depth error based on the ground truth, and provide a measure of RMSE of points across the plane surface.

For depth characterisation, the Intel RealSense D456 was mounted to a tripod, and pointed at a white wall. This was done instead of using the hardware rig since the mounting brackets failed to keep the camera parallel to the floor. The mount would result in the camera's angle to the ground varying between heights, and the angle would consistently remain above 1.5 degrees. This angle offset and variation would pollute the results of the distance sweep, and thus a vertical scan against a white wall was used. The camera was moved across a range of distance from 600mm to 900mm, and the camera to plane angle was manually minimised at each distance.

For angle characterisation, the camera was mounted to the experimental rig and angles ranging from $-25°$ to $+25°$ at $2°$ intervals were taken. The experimental rig was used as this would allow for easy control of the camera angle using the servo motor built into the mount. A large flat white foam board was placed on the floor, and was continuously adjusted to remain in the centre of the cameras field of view (FOV) across different angles. The camera was set to a fixed height, and the same custom depth quality code was used to scan the flat surface at different heights. The camera was rotated about two axes independently, one parallel to the axis going through the IR stereo camera pair which will be called the pitch axis, and the second axis running perpendicular to both the pitch and axial axis, this will be called the roll axis. At each angle, the vertical z distance from the left IR camera lens to the floor panel was measured, using this measurement and the measured angle, the

code would determine the axial distance from the optical origin to the centre of the plane, and use this as the ground truth for the plane fit.

For light intensity measurement, scans of the same foam board was taken under a variety of lighting conditions. Most of the experimental testing occurred indoors during the afternoon. with a range from open blinds and lights on to the night with no lights and blinds closed. Additional testing occurred at mid day in direct sunlight and shaded sunlight to obtain extremities of lighting conditions. The cameras exposure was left to automatic, and the same code was used to measure error statistics across the planar fit.

These characterisations would provide a preliminary understanding of the general performance across different hardware and environmental conditions, and would be used to inform decisions made for later characterisation.

5.2 3D Reference Characterisation

once the planar characterisation was performed, rigorous measurements were made using the reference objects mentioned earlier. A modified version of the main capture program was used to capture scans of the different reference objects under different software settings, and different hardware set-ups.

Firstly, scans of the large sinusoidal surface, which will be called the large egg box shape, and large single pyramid were taken under fixed a fixed height and angle, but with ranging camera configurations were taken to determine the optimal configuration settings. These scans ranged from resolutions spanning from 1280x720 to 640x320, depth units ranging from 0.001m to 0.0001m, and across all pre-sets excluding hand.

Next, the camera was configured to use a resolution of 1280x720, depth units of 0.0001mm and a high density preset. Then several scans were taken of the large egg box, and small eggbox were taken at different heights and angles.

At each height, 5 stationary scans of each shape were taken without tilt or rotation. These would then be used to iterate through different combinations of filter settings provided by the RealSense SDK in post processing. This would provide insight into the optimal filter settings across the different heights.

Then subsequently, a full scan of each shape was taken, at rotations $-45°$ to $90°$ at $45°$ increments, and tilts from $-25°$ to $25°$ at $5°$ increments. This would allow for analysis of reconstruction performance across different combinations of tilt and rotation fusion in post processing.

5.3 Considerations???

Chapter 6

Data Processing Pipeline

6.1 Depth Image Cropping

6.2 Depth Image Fusion

6.3 Surface Rugosity Estimation

6.4 Feature Detection

Chapter 7

Experimental Validation

7.1 Known Relief and Controlled Targets

7.2 Roughness Metric Validation

Chapter 8

Results

8.1 Characterisation Results

8.2 Roughness Metric Results

8.3 Feature Segmentation Results

Chapter 9

Discussion

9.1 Optimal Configuration

9.2 Roughness Estimation

9.3 Feature Detection

Chapter 10

Conclusions

Chapter 11

Further work

11.1 Characterisation

11.2 Hardware Choices

Appendix A

GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year project report can be found.

A.1 Title for appendix A.1

- GitHub Repository: Repo name

Appendix B

Reference Objects

testing something

Appendix B: Ethics approval

Appendix C: Topic description

The original topic description used during the project selection process is included below. This topic has since been further refined, and the exact problem that was investigated is described in Section 1.2.

Appendix D: ECSA Graduate Attributes

Project mapping to ECSA Graduate Attributes

The table below provides a structured reflection on where each of the required Graduate Attributes (GAs) as defined by Engineering Council of South Africa (ECSA) associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

GA	Graduate Attribute Description	Student reflection: How this outcome was achieved during the project
GA1	*Problem solving*	
GA4	*Investigations, experiments and data analysis*	
GA5	*Use of engineering tools*	
GA6	*Professional and technical communication*	
GA8	*Individual and collaborative teamwork*	
GA9	*Independent learning ability*	

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